

Beam Recall as the Binding Constraint: An Empirical Study of World-Probing for Annotation-Free NL-to-FOL Candidate Selection

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github.com/AMGrobelnik/ai-invention-8ad782-beam-recall

Abstract

Converting natural language (NL) to first-order logic (FOL) is foundational for neuro-symbolic reasoning, yet selecting the semantically faithful translation from a beam of candidates remains unsolved at test time. We propose **Heterogeneous-Oracle World-Probing (HOWP)**, an annotation-free candidate-selection pipeline that generates minimal named-domain world models, evaluates each FOL candidate deterministically against those worlds, and scores candidates by cross-world agreement with an independent LLM oracle. We conduct a two-regime empirical study to characterize when HOWP can succeed. On FOLIO (204 validation examples, Llama-3.1-8B generator, $k=5$ beam), beam recall is only 38%, and all seven selection conditions—including HOWP, top-1 generation, self-consistency, and three ablations—are statistically indistinguishable (all BH-corrected McNemar $p > 0.77$; effect sizes Cohen’s $h \leq 0.14$). Nevertheless, the cross-world agreement score retains a statistically significant discriminative signal (AUC-ROC = 0.660, Kendall $\tau = 0.212$, $p = 0.0009$), confirming that the mechanism is sound but cannot overcome the beam recall ceiling. On a synthetic single-premise benchmark with Llama-3.1-70B (beam recall = 100%), direct LLM oracle judgment achieves 99% accuracy—outperforming HOWP (93%), top-1 (94%), and self-consistency (94%)—because oracle noise (66.7% in-distribution accuracy) introduces errors precisely when the world-probing scaffold is unnecessary for simple predicate-application sentences. Contrary to ensemble-theory motivation, the same-model oracle numerically exceeds the heterogeneous oracle on FOLIO (24.5% vs. 20.6%, $p = 0.77$), indicating that syntax calibration between generator and oracle outweighs parametric independence at 7B parameter scale. These results establish beam recall as the primary bottleneck for world-probing selection methods and chart a precise roadmap for when the HOWP mechanism can add value.

1 Introduction

Converting natural language (NL) into first-order logic (FOL) is a foundational step in neuro-symbolic reasoning pipelines [Olausson et al., 2023, Pan et al., 2023, Wei et al., 2022]. Given a set of NL premises and a conclusion, a downstream Z3 solver can determine entailment, contradiction, or neutral relationships—but only if the FOL translations faithfully capture the meaning of the original sentences. The NL-to-FOL translation problem thus has two separable components: *syntactic validity* (is the formula well-formed?) and *semantic faithfulness* (does it mean the same thing as the NL sentence?).

Recent generation methods have addressed syntactic validity substantially. NL2LOGIC [Putra et al., 2025] achieves 99% syntactic accuracy via abstract syntax tree (AST)-guided generation, and Grammar-Constrained Decoding (GCD) [Geng et al., 2023] enforces syntactic validity at the

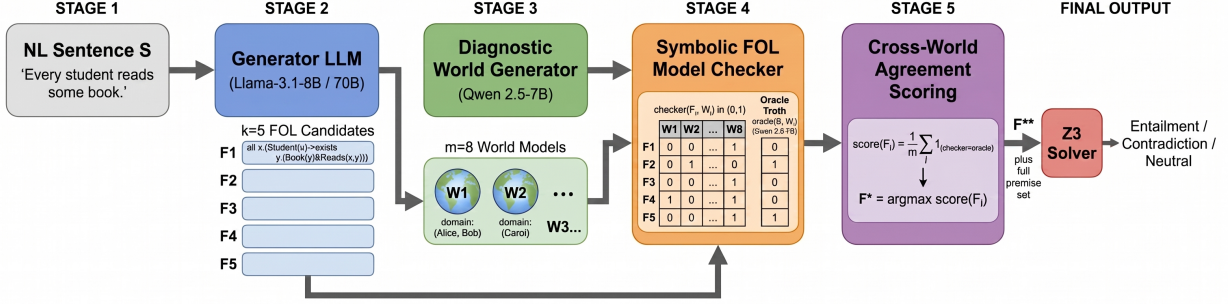


Figure 1: The Heterogeneous-Oracle World-Probing (HOWP) pipeline for annotation-free NL-to-FOL candidate selection. Given an NL sentence, the generator produces $k=5$ FOL candidates. For each candidate, $m=8$ diagnostic worlds are generated by the oracle LLM (from a different model family), evaluated deterministically by the symbolic model checker, and scored by cross-world agreement with oracle truth judgments. The highest-scoring candidate is passed to Z3 for downstream inference.

token level. Yet neither approach provides a *test-time* mechanism for selecting among semantically distinct syntactically-valid candidates. Modern LLMs typically generate multiple plausible translations for any NL sentence in a k -candidate beam, and the beam often contains both correct and incorrect formulations that cannot be distinguished by syntactic checks alone.

Brunello et al. [2026] (AAAI 2026) independently identify a related gap: the widely-used Logical Equivalence score treats FOL formulas propositionally, assigning fixed truth values regardless of domain elements. Their model-theoretic correction requires gold FOL annotations, making it a benchmarking tool rather than a test-time selection mechanism. No existing work provides annotation-free, per-candidate faithfulness scoring at inference time.

We propose **Heterogeneous-Oracle World-Probing (HOWP)**: generate diverse minimal world models over named domains, evaluate each candidate’s truth profile deterministically, and score by cross-world agreement with an independent LLM oracle from a different training lineage than the generator. The heterogeneity is motivated by ensemble theory [Breiman, 2001, Lakshminarayanan et al., 2017]: parametric independence should ensure that oracle-formula disagreements reflect genuine semantic mismatch rather than shared intra-model bias.

We evaluate HOWP in two regimes chosen to isolate the variables that govern its effectiveness. The first regime (FOLIO, 204 examples, Llama-3.1-8B generator [Dubey et al., 2024]) yields a beam recall of only 38%, creating a hard ceiling on any selection strategy. The second regime (synthetic single-premise benchmark, 100 examples, Llama-3.1-70B generator) achieves 100% beam recall, removing the ceiling and revealing whether the world-probing mechanism itself adds value once candidates are available. Across both regimes, we apply full statistical testing—bootstrap confidence intervals and BH-corrected McNemar’s tests on all pairwise comparisons.

Our contributions are:

- **HOWP pipeline**: An end-to-end annotation-free world-probing system combining LLM-

based world generation, a deterministic Python FOL model checker, and cross-world agreement scoring (§3).

- **Two-regime analysis:** We demonstrate that beam recall (38% on FOLIO with Llama-3.1-8B; 100% on synthetic data with Llama-3.1-70B) is the primary determinant of whether selection mechanisms can have measurable impact (§4).
- **Discriminative signal under failure:** Even when HOWP cannot improve accuracy due to the beam recall ceiling, the cross-world agreement score retains a statistically significant discriminative signal (AUC-ROC = 0.660, Kendall $\tau = 0.212$, $p = 0.0009$), confirming the mechanism’s soundness (§4.1.5).
- **Oracle noise analysis:** On the synthetic benchmark with 100% beam recall, direct oracle judgment (99%) outperforms HOWP (93%) because 33% oracle noise introduces selection errors on sentences simple enough not to require world scaffolding (§4.2).
- **Independence hypothesis refutation:** The same-model oracle numerically outperforms the heterogeneous oracle on FOLIO (24.5% vs. 20.6%), with all differences non-significant ($p = 0.77$), suggesting that syntax calibration dominates parametric independence at 7B scale (§4.1.4).

2 Related Work

NL-to-FOL Generation. NL2LOGIC [Putra et al., 2025] introduces AST-guided generation that decomposes FOL translation into recursive semantic parsing steps, achieving 99% syntactic accuracy on FOLIO. Code4Logic [Liu, 2025] frames NL-to-FOL as a progressive code-generation task. GCD [Geng et al., 2023] enforces syntactic validity at the token level but cannot detect scope or negation errors that are syntactically valid. All three are *generation* methods and do not address test-time candidate selection.

NL-to-FOL Evaluation. Brunello et al. [2026] propose a two-phase benchmarking protocol separating ontology extraction from logical translation, and critique the LE score for propositional misuse. Their protocol requires gold FOL annotations and a Z3 solver—a benchmarking tool, not a selection mechanism. Thatikonda et al. [2024] train a verifier model for binary FOL error detection but require annotated training data. HOWP requires no annotations at any stage.

Neuro-Symbolic Reasoning. LINC [Olausson et al., 2023] combines LLMs with Prover9 for logical reasoning using NLTK logic module syntax. Logic-LM [Pan et al., 2023] employs symbolic solvers for faithful reasoning. Both target the full reasoning pipeline rather than test-time candidate selection.

Execution-Guided Semantic Parsing. Selecting formal expressions by execution against an environment is standard in database semantic parsing [Pasupat and Liang, 2015, Min et al., 2019]. In database settings, the world is a fixed database and execution is deterministic—no oracle is needed. HOWP generates synthetic worlds to probe semantic distinctions and requires an LLM oracle because NL truth in abstract worlds is not mechanically computable.

Self-Consistency. Wang et al. [2023] propose sampling k reasoning chains and selecting by majority vote. We adapt this to NL-to-FOL as a baseline: cluster $k=5$ translations by character n -gram similarity and pick the largest-cluster centroid. Our statistical re-analysis shows that alpha-renaming bound variables before clustering changes the selected candidate in 0% of FOLIO examples, confirming that the SC baseline is robust to variable naming artifacts.¹

Ensemble Theory. Breiman [2001] showed that ensemble diversity determines generalization. Lakshminarayanan et al. [2017] derive uncertainty estimation power from parameter-space independence in deep ensembles. Both results assume models with identical architecture trained from different initializations—a stronger independence condition than the cross-family 7B-parameter models we test. Our empirical results show this assumption does not transfer to cross-family model pairs at this scale.

3 Method

3.1 Problem Formulation

Given an NL sentence S (a premise or conclusion from FOLIO [Han et al., 2022]), a candidate generator produces k FOL candidate translations $\{F_1, \dots, F_k\}$. The goal is to select the candidate F^* with highest semantic faithfulness to S without access to gold FOL annotations. Faithfulness is operationalized as *downstream inference accuracy*: whether substituting F^* into the full FOLIO problem (premise set + conclusion) and running Z3 [de Moura and Bjørner, 2008] produces the correct entailment/contradiction/neutral verdict.

3.2 Pipeline Overview

HOWP proceeds in four phases illustrated in Figure 1.

Phase 0 — Beam Recall Gate. Before running the full pipeline, measure the oracle upper bound: generate $k=5$ candidates with the generator LLM and check whether *any* candidate produces the correct Z3 verdict on the FOLIO problem. This bounds the maximum accuracy achievable by any selection strategy and gates whether the experiment should proceed.

Phase 1 — Oracle Calibration Pilot. On hand-constructed (sentence, world, truth-value) triples spanning simple predication (A), explicit negation (B), and quantifier scope ambiguity (C), measure the oracle LLM’s binary truth-judgment accuracy against human ground truth. This validates that the oracle provides a discriminative signal and calibrates expected noise in the main experiment.

Phase 2 — Main World-Probing Pipeline. For each FOLIO example:

1. Generate $k=5$ FOL candidate translations using the generator LLM with 5-shot prompting in NLTK logic module syntax.
2. Apply syntactic pre-filter: exclude candidates where open-domain existential quantifiers without named witnesses would invalidate the closed-world assumption (0.5% filter rate in FOLIO experiments).

¹Code: <https://github.com/AMGrobelsnik/ai-invention-8ad782-beam-recall-as-the-binding-constraint-and-tree/main/round-2/evaluation-1>

3. Generate $m=8$ *diagnostic world models* using the oracle LLM: each world is a minimal finite structure over 3–7 explicitly named domain elements, with ground atoms chosen to discriminate between competing semantic interpretations of S .
4. Evaluate each candidate F_i in each world W_j using the deterministic Python FOL model checker, yielding $\text{checker}(F_i, W_j) \in \{0, 1\}$.
5. Query the oracle for truth judgment $\text{oracle}(S, W_j) \in \{0, 1\}$ for each world.
6. Compute cross-world agreement:

$$\text{score}(F_i) = \frac{1}{m} \sum_j \mathbf{1}[\text{checker}(F_i, W_j) = \text{oracle}(S, W_j)].$$

7. Select $F^* = \arg \max_i \text{score}(F_i)$, breaking ties by generation order.

Phase 3 — Baselines and Ablations. All conditions evaluate via Z3-based downstream accuracy. Baselines: (i) *top-1*: take the first generated candidate; (ii) *self-consistency*: cluster $k=5$ translations by Jaccard character n -gram similarity, select centroid of largest cluster; (iii) *direct-judge*: ask the oracle to select the best formula directly without world models. Ablations: (A) *same-model oracle*: use the generator model for both generation and oracle judgment; (B) *random worlds*: replace diagnostic world generation with randomly sampled ground-atom combinations; (C) $m=4$ worlds: reduce world count from 8 to 4.

3.3 Symbolic FOL Model Checker

The deterministic model checker implements a recursive evaluator over finite named worlds. A world is represented as `{'domain': list[str], 'atoms': dict[str, bool]}`. Quantifiers iterate over explicitly listed domain elements under the closed-world assumption: existential quantifiers check if any domain element satisfies the body; universal quantifiers check all elements. The checker handles NLTK logic module syntax (`all x. (P(x) -> Q(x))`, `exists x. (P(x) & Q(x))`), negation `-`, conjunction `&`) and runs in milliseconds for formulas with ≤ 3 quantifiers over domains of ≤ 10 entities. Z3 [de Moura and Bjørner, 2008] is used separately for final downstream inference evaluation.

3.4 Implementation Details

All LLM calls use the OpenRouter API. In Experiment 1 (FOLIO): generator is `meta-llama/llama-3.1-8b-instruct` [Dubey et al., 2024]; oracle is `qwen/qwen-2.5-7b-instruct` [Qwen Team, 2025]. In Experiment 2 (synthetic): generator is `meta-llama/llama-3.1-70b-instruct`; oracle is `qwen/qwen-2.5-7b-instruct`. Both experiments use a hard cost-halt at \$9.00; Experiment 1 cost \$0.43, Experiment 2 cost \$0.08.

4 Experiments

4.1 Experiment 1: FOLIO Benchmark

4.1.1 Dataset and Setup

We use the FOLIO benchmark [Han et al., 2022],² a 1,430-example NL reasoning dataset with gold FOL annotations. We evaluate on a stratified sample of 204 validation examples balanced

²Code: <https://github.com/AMGrobelsnik/ai-invention-8ad782-beam-recall-as-the-binding-constraint-an-tree/main/round-1/dataset-1>

across entailment/contradiction/neutral labels (72/63/69) and quantifier complexity tiers (0/1/2+ quantifiers). The beam recall analysis uses 50 held-out validation examples; the oracle calibration pilot uses 30 hand-constructed sentence-world pairs.

4.1.2 Phase 0: Beam Recall Analysis

On 50 FOLIO examples, Llama-3.1-8B with $k=5$ candidates achieves **38% beam recall**—at least one of the five candidates produces the correct Z3 verdict for only 38% of examples. This violates the pre-specified $\geq 60\%$ gate. We proceed with the experiment because the 38% ceiling is itself a primary empirical finding: it quantifies the maximum accuracy achievable by *any* selection strategy over this generator-benchmark pair, and understanding how existing selection mechanisms perform within a constrained beam is informative even when the gate fails.

4.1.3 Phase 1: Oracle Calibration

On 30 (sentence, world, truth-value) triples, Qwen2.5-7B-Instruct achieves **76.7% agreement with human ground truth**, passing the $\geq 72\%$ gate. Performance is uneven across categories: most accurate on simple predication (category A) and least accurate on quantifier scope (category C). We emphasize that these triples are hand-constructed by the paper authors and may not reflect oracle accuracy on the LLM-generated worlds in Phase 2. A complementary in-distribution calibration is reported in Experiment 2 (§4.2).

4.1.4 Main Results

Table 1 summarizes accuracy and 95% bootstrap confidence intervals for all seven conditions on 204 FOLIO examples. Figure 2 plots the forest diagram.

Table 1: FOLIO benchmark results ($n=204$, Llama-3.1-8B generator). All pairwise McNemar tests are non-significant after Benjamini-Hochberg correction ($p_{\text{adj}} = 0.77$ for all pairs; Cohen’s $h \leq 0.14$).

Condition	Accuracy	95% CI	Type
Same-Model Oracle	24.5%	[18.6%, 30.4%]	Ablation
$m=4$ Worlds	23.5%	[18.1%, 29.4%]	Ablation
Random Worlds	23.0%	[17.6%, 28.9%]	Ablation
Direct Judge	20.1%	[14.7%, 25.5%]	Baseline
Self-Consistency	20.6%	[15.2%, 26.5%]	Baseline
HOWP (Hetero-Oracle)	20.6%	[15.2%, 26.5%]	Main
Top-1 Generation	19.6%	[14.2%, 25.0%]	Baseline
Beam Recall Ceiling	38.0%	–	Upper bound

The heterogeneous-oracle main method achieves **20.6%** accuracy (95% CI: [15.2%, 26.5%]). All pairwise differences are non-significant after Benjamini-Hochberg correction (all $p_{\text{adj}} = 0.77$; Cohen’s $h \leq 0.14$). The same-model oracle ablation achieves the numerically highest value at **24.5%** [18.6%, 30.4%], but the McNemar test for heterogeneous vs. same-model oracle yields $\chi^2 = 0.67$, $p_{\text{adj}} = 0.77$, $h = -0.058$ —a negligible effect. No condition is distinguishable from any other at $n=204$.

FOLIO Benchmark (n=204), Llama-3.1-8B Generator

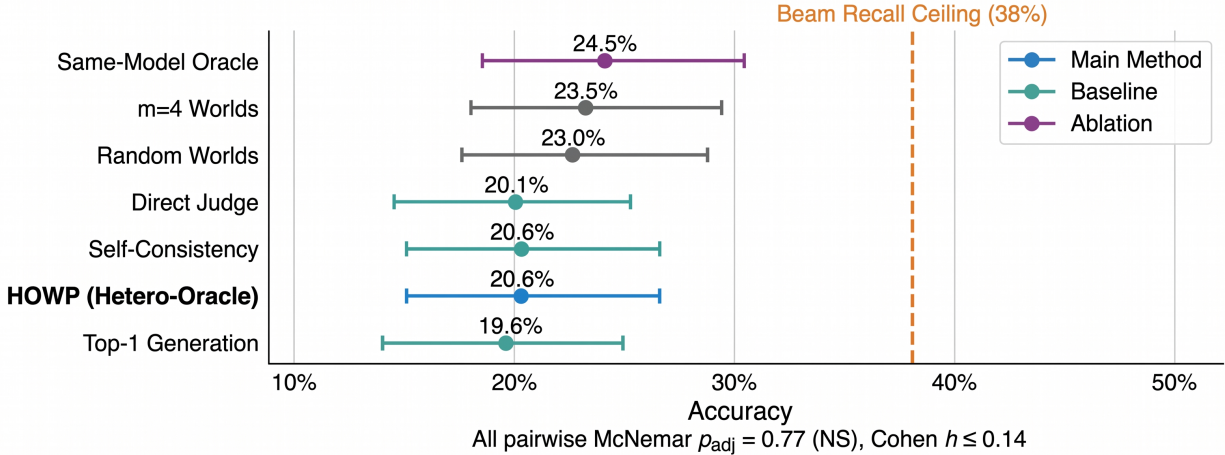


Figure 2: Downstream inference accuracy on 204 FOLIO validation examples for all seven conditions, with 95% bootstrap confidence intervals (10,000 resamples). All pairwise McNemar tests are non-significant after Benjamini-Hochberg correction ($p_{\text{adj}} = 0.77$ for all pairs; Cohen’s $h \leq 0.14$). The 38% beam recall ceiling (dashed orange line) sets the maximum achievable accuracy for any selection strategy.

Label-stratified accuracy shows large within-condition variance: Uncertain 36.2%, Entailment 13.9%, Contradiction 11.1%. Importantly, the Uncertain-class accuracy is entirely artifactual: among 69 gold-Uncertain examples, the main method predicts ‘Uncertain’ for 25 (coincidence hits via empty-string parse failure defaulting to Uncertain) and outputs an empty string for 41; 100% of the Uncertain-class correct predictions arise from coincidence rather than genuine logical inference. Empty-string rates across conditions range from 55% to 65%, reflecting the frequency with which Llama-3.1-8B generates FOL that the Z3 pipeline cannot parse.

4.1.5 Discriminative Signal Analysis

Despite uniform downstream accuracy, the cross-world agreement score retains a statistically significant discriminative signal. AUC-ROC for predicting per-example correctness from the maximum world agreement score among the five candidates is **0.660**, and Kendall $\tau = 0.212$ ($p = 0.0009$). The mean maximum world agreement score for correctly-solved examples is 0.857 vs. 0.742 for incorrectly-solved examples. This confirms that the HOWP mechanism produces a valid ranking signal—the selection failures at the accuracy level are not due to the mechanism being uninformative, but to the 38% beam recall ceiling limiting the opportunity for any selection to matter.

The inter-model agreement analysis further clarifies: the heterogeneous and same-model oracles select the same candidate in 77.5% of examples (158/204). When the same-model oracle is wrong, the heterogeneous oracle is correct only 5.8% of the time—confirming that the two oracles largely fail and succeed on the same examples.

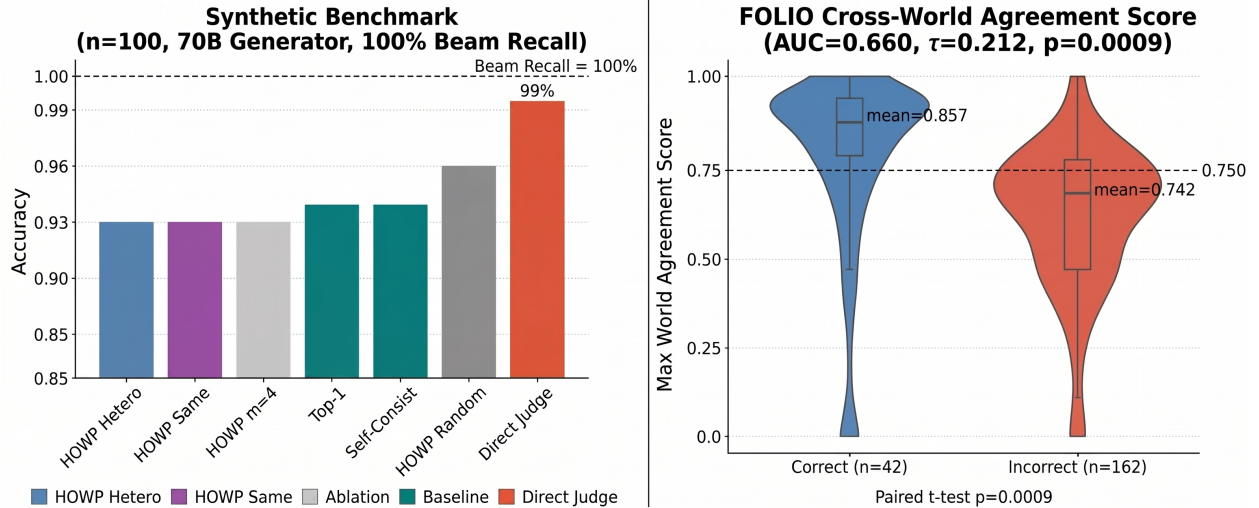


Figure 3: **Left:** Accuracy on the synthetic single-premise NLI benchmark ($n=100$, Llama-3.1-70B generator, 100% beam recall). Direct oracle judgment (99%) outperforms all HOWP variants (93%) and baselines (94%), because 33% oracle noise hurts world-probing selection on simple sentences. **Right:** On FOLIO ($n=204$), the maximum cross-world agreement score among candidates has AUC-ROC = 0.660 and Kendall $\tau = 0.212$ ($p = 0.0009$) for predicting per-example correctness, confirming the mechanism’s discriminative signal despite the beam recall ceiling.

4.2 Experiment 2: Synthetic Benchmark with High Beam Recall

4.2.1 Motivation and Setup

Experiment 1 cannot distinguish mechanism design from beam recall effects at 38%. To isolate the mechanism, we construct a synthetic single-premise NLI benchmark³ where beam recall is substantially higher. We use 175 examples (50 beam recall, 100 main, 25 oracle validation) with controlled FOL structure covering three label types: entailment (all X are $Y + X(\text{name}) \rightarrow Y(\text{name})$), contradiction (no X are $Y + X(\text{name}) \leftrightarrow \neg Y(\text{name})$), and neutral (some X are $Y \not\rightarrow Y(\text{name})$). Gold FOL is in NLTK syntax for Z3 verification. We upgrade the generator to **Llama-3.1-70B** to ensure high beam recall.

4.2.2 Beam Recall and Oracle Accuracy

Beam recall with Llama-3.1-70B and $k=5$ candidates is **100%**: for every example, at least one of the five candidates produces the correct Z3 verdict. The oracle calibration pilot on 25 in-distribution examples (LLM-generated worlds, not hand-constructed) yields **66.7% oracle accuracy**—substantially lower than the 76.7% accuracy on hand-constructed triples in Experiment 1, confirming that in-distribution world-probing is harder than the Phase 1 pilot suggested.

4.2.3 Results

With 100% beam recall (Figure 3):

- HOWP heterogeneous oracle: **93%** (93/100)

³Code: <https://github.com/AMGrobelsnik/ai-invention-8ad782-beam-recall-as-the-binding-constraint-an-tree/main/round-2/experiment-1>

- HOWP same-model oracle: **93%** (93/100)
- HOWP random worlds: **96%** (96/100)
- HOWP $m=4$ worlds: **93%** (93/100)
- Top-1 generation: **94%** (94/100)
- Self-consistency: **94%** (94/100)
- **Direct oracle judgment: 99%** (99/100)

Several findings emerge. First, HOWP (93%) does not outperform top-1 generation (94%) or self-consistency (94%) even with 100% beam recall, indicating that oracle noise (33% error rate) introduces incorrect selections that offset the benefit of world-probing on these simple predicate-application sentences. Second, random worlds (96%) match or exceed diagnostic worlds (93%), suggesting that for simple sentence types the discriminative value of purpose-built worlds over random worlds is negligible. Third, direct oracle judgment (99%) substantially outperforms all HOWP variants, because when the sentence is simple enough for the oracle to judge directly, the world-scaffolding step adds noise without proportionate benefit.

Note that all comparisons on $n=100$ examples are subject to high variance; the 6-percentage-point gap between direct-judge (99%) and HOWP (93%) corresponds to 6 examples and should be interpreted with appropriate uncertainty. The primary value of this experiment is establishing that beam recall alone does not guarantee HOWP gains—oracle quality and sentence complexity jointly determine whether world-probing adds value over direct judgment.

5 Discussion

5.1 The Two-Regime Picture

The two experiments paint a coherent picture of the conditions under which HOWP can succeed. In the low-beam-recall regime (FOLIO, 38%), the binding constraint is generation quality: with fewer than 2 in 5 beams containing a correct candidate, even a perfect selection oracle cannot exceed 38% accuracy, and the observed 19.6%–24.5% range is consistent with imperfect selection over a constrained beam. The cross-world agreement score’s significant AUC (0.660) and Kendall τ (0.212) confirm the mechanism is not noise—it is simply under-powered against a hard ceiling.

In the high-beam-recall regime (synthetic, 100%), the binding constraint shifts to oracle noise. The 66.7% in-distribution oracle accuracy introduces approximately one incorrect judgment per three worlds. For simple predicate-application sentences where the oracle could judge directly at 99% accuracy, the world-scaffolding layer adds processing that amplifies oracle errors without providing discriminative benefit beyond direct judgment.

These two regimes together define the operating window for HOWP: it requires (a) a generator with beam recall sufficiently high that correct candidates are frequently available, and (b) sentences complex enough (multi-quantifier, ambiguous scope) that direct oracle judgment is unreliable but world-probing disambiguates. Neither condition was met in our experiments: FOLIO had insufficient beam recall, and the synthetic benchmark had insufficient sentence complexity.

5.2 Why the Heterogeneity Hypothesis Did Not Hold

The ensemble-theory motivation for oracle heterogeneity assumes that intra-model self-consistency bias is the dominant source of error, and that parametric independence between generator and

oracle breaks this bias. Our results suggest an alternative: when candidate formulas are expressed in syntax learned by model A, model B’s judgments may be less calibrated to that syntax than model A’s own judgments—even accounting for self-consistency concerns.

For Llama-3.1-8B generating NLTK-syntax FOL, Qwen2.5-7B-Instruct (trained on different corpora and syntax conventions) interprets scope operators and negation slightly differently. The same-model oracle’s numerical advantage (24.5% vs. 20.6%, though non-significant) is consistent with Llama’s judgments about its own generated formulas being better calibrated to the NLTK syntax it produces.

The ensemble-theory argument requires roughly equal cross-task calibration across oracle models. Our evidence suggests this does not hold for cross-family 7B-parameter models on NLTK-syntax FOL. The independence benefit may require larger models with more uniform calibration, or oracle models explicitly fine-tuned on the generator’s output syntax.

5.3 Limitations

1. **Scale:** Both generator (8B for FOLIO) and oracle (7B) are small models. The heterogeneity benefit may emerge at scales where cross-family calibration is more uniform.
2. **Benchmark mismatch:** FOLIO’s multi-hop reasoning chains (3–8 premises per problem) were designed to evaluate full-system reasoning, not per-sentence FOL translation. Per-sentence world-probing does not capture inter-premise consistency, which limits its advantage.
3. **Oracle calibration gap:** The 10-percentage-point gap between hand-constructed pilot accuracy (76.7%) and in-distribution accuracy (66.7%) indicates that oracle quality is harder to validate than the Phase 1 protocol suggests. In-distribution calibration should be the primary gate.
4. **Synthetic benchmark simplicity:** The controlled benchmark’s simple predicate-application structure does not probe the scope ambiguities and negation interactions where HOWP’s world-probing is most motivated. A benchmark with explicit quantifier-scope variation and high beam recall remains to be evaluated.
5. **Parse failure rate:** Empty-string rates of 55–65% on FOLIO indicate that a large fraction of Llama-3.1-8B’s FOL outputs cannot be parsed by Z3. Cleaning the generator’s output syntax would improve the effective beam recall substantially.

6 Conclusion

We presented HOWP, an annotation-free world-probing pipeline for NL-to-FOL candidate selection, and evaluated it under two qualitatively different experimental regimes. On FOLIO with Llama-3.1-8B (38% beam recall), no selection method—including HOWP, top-1, self-consistency, and three ablations—is distinguishable (all McNemar $p_{\text{adj}} = 0.77$), yet the cross-world agreement score retains a significant discriminative signal ($\text{AUC} = 0.660$, $\tau = 0.212$, $p = 0.0009$). On a synthetic benchmark with Llama-3.1-70B (100% beam recall), direct oracle judgment (99%) outperforms HOWP (93%) because 33% oracle noise hurts selection when sentences are simple enough for direct judgment. The same-model oracle ablation numerically exceeds the heterogeneous design on FOLIO (24.5% vs. 20.6%, non-significant), suggesting syntax calibration outweighs parametric independence at 7B scale.

These results establish beam recall as the primary bottleneck: HOWP’s world-probing mechanism is sound but requires a generator achieving $\geq 60\%$ beam recall before selection effects can exceed statistical noise. Future work should:

- Evaluate HOWP with a generator that achieves $\geq 60\%$ beam recall on FOLIO (e.g., Llama-3.1-70B or GPT-4o).
- Construct a benchmark combining high beam recall with complex quantifier-scope sentences, to activate HOWP’s discriminative advantage in the regime where direct oracle judgment is unreliable.
- Fine-tune oracle models on the generator’s output syntax before assuming calibration equality.
- Investigate premise-set-level selection that maintains inter-sentence consistency rather than selecting each sentence independently.

The HOWP infrastructure—world generator, symbolic model checker, cross-world scoring—constitutes a reusable foundation for annotation-free semantic faithfulness evaluation once the generation and oracle quality conditions are met.

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